

1. Integer Partitions

1(a) Find $p(6)$

Through brute force, we find all partitions of 6:

$$\begin{aligned} &6 \\ &5 + 1 \\ &4 + 2 \\ &4 + 1 + 1 \\ &3 + 3 \\ &3 + 2 + 1 \\ &3 + 1 + 1 + 1 \\ &2 + 2 + 2 \\ &2 + 2 + 1 + 1 \\ &2 + 1 + 1 + 1 + 1 \\ &1 + 1 + 1 + 1 + 1 + 1 \end{aligned}$$

Therefore, by counting, we have:

$$p(6) = 11$$

1(b) $p_{\leq 3}(6)$ and partitions with parts of size ≤ 3

Through brute force, with restrictions, we find all partitions of 6 into **at most 3 parts**:

$$\begin{aligned} &6 \\ &5 + 1 \\ &4 + 2 \\ &4 + 1 + 1 \\ &3 + 3 \\ &3 + 2 + 1 \\ &2 + 2 + 2 \end{aligned}$$

Therefore, by counting, we have:

$$p_{\leq 3}(6) = 7$$

We brute force (again) the partition of 6 into parts of size **at most 3**:

$$\begin{aligned}
& 3 + 3 \\
& 3 + 2 + 1 \\
& 3 + 1 + 1 + 1 \\
& 2 + 2 + 2 \\
& 2 + 2 + 1 + 1 \\
& 2 + 1 + 1 + 1 + 1 \\
& 1 + 1 + 1 + 1 + 1 + 1
\end{aligned}$$

And so, there are also 7.

1(c) $p_3(6)$ and $p_{\leq 3}(3)$

Via brute force, we find the partition of 6 into **exactly 3 parts**:

$$\begin{aligned}
& 4 + 1 + 1 \\
& 3 + 2 + 1 \\
& 2 + 2 + 2
\end{aligned}$$

Therefore, we have:

$$p_3(6) = 3$$

And finally, we find partitions of 3 into **at most 3 parts**:

$$\begin{aligned}
& 3 \\
& 2 + 1 \\
& 1 + 1 + 1
\end{aligned}$$

And, last but not least, we have the following:

$$p_{\leq 3}(3) = 3$$

2. Generating Functions

We are given the following:

$$\prod_{k=1}^6 \frac{1}{1-x^k} = 1 + x + 2x^2 + 3x^3 + 5x^4 + 7x^5 + 11x^6 + 14x^7 + 20x^8 + 26x^9 + 35x^{10} + \dots$$

2(a) Coefficients of $\prod_{k=1}^7 \frac{1}{1-x^k}$

$$\prod_{k=1}^7 \frac{1}{1-x^k} = \left(\prod_{k=1}^6 \frac{1}{1-x^k} \right) (1 + x^7 + x^{14} + \dots)$$

For powers up to x^{10} , we find we only need to add the x^7 shift.

Therefore, the coefficients become:

$$\begin{aligned} &1, 1, 2, 3, 5, 7, 11, \\ &14 + 1 = 15, \\ &20 + 1 = 21, \\ &26 + 2 = 28, \\ &35 + 3 = 38 \end{aligned}$$

Finally, we have:

$$1 + x + 2x^2 + 3x^3 + 5x^4 + 7x^5 + 11x^6 + 15x^7 + 21x^8 + 28x^9 + 38x^{10}$$

2(b) Find $p_{\leq 7}(8)$

Trivially, from the coefficient above:

$$p_{\leq 7}(8) = 21$$

3. Partitions into Powers of 2

Let $q(k)$ be the number of partitions of k into powers of 2, with $q(0) = 1$.

We observe that the allowed parts are

$$1, 2, 4, 8, 16, \dots$$

Similarly, for each power of 2, we may use it any number of times. Therefore, we have:

- $1 = 2^0$

$$1 + x + x^2 + x^3 + \dots = \frac{1}{1-x},$$

- $2 = 2^1$

$$1 + x^2 + x^4 + x^6 + \dots = \frac{1}{1-x^2},$$

- $4 = 2^2$

$$1 + x^4 + x^8 + x^{12} + \dots = \frac{1}{1-x^4},$$

- 2^i

$$\frac{1}{1-x^{2^i}}.$$

From lecture, we observe multiplying these together gives the ordinary generating function for $(q(k))_{k \geq 0}$:

$$\sum_{k \geq 0} q(k)x^k = \prod_{i=0}^{\infty} \frac{1}{1 - x^{2^i}}$$

Hence the ordinary generating function is

$$\sum_{k \geq 0} q(k)x^k = \prod_{i=0}^{\infty} \frac{1}{1 - x^{2^i}}.$$

4. Partitions into Distinct Parts

We observe that each integer i can appear either 0 or 1 time. Therefore, the generating function for part i is

$$1 + x^i$$

Multiplying for all i gives

$$\sum_{n \geq 0} q(n)x^n = \prod_{i=1}^{\infty} (1 + x^i)$$

where $q(n)$ counts partitions of n into distinct parts. Using the example in the problem,

$$10 = 5 + 3 + 2$$

is allowed, but

$$5 + 3 + 1 + 1$$

is not because the part 1 repeats.